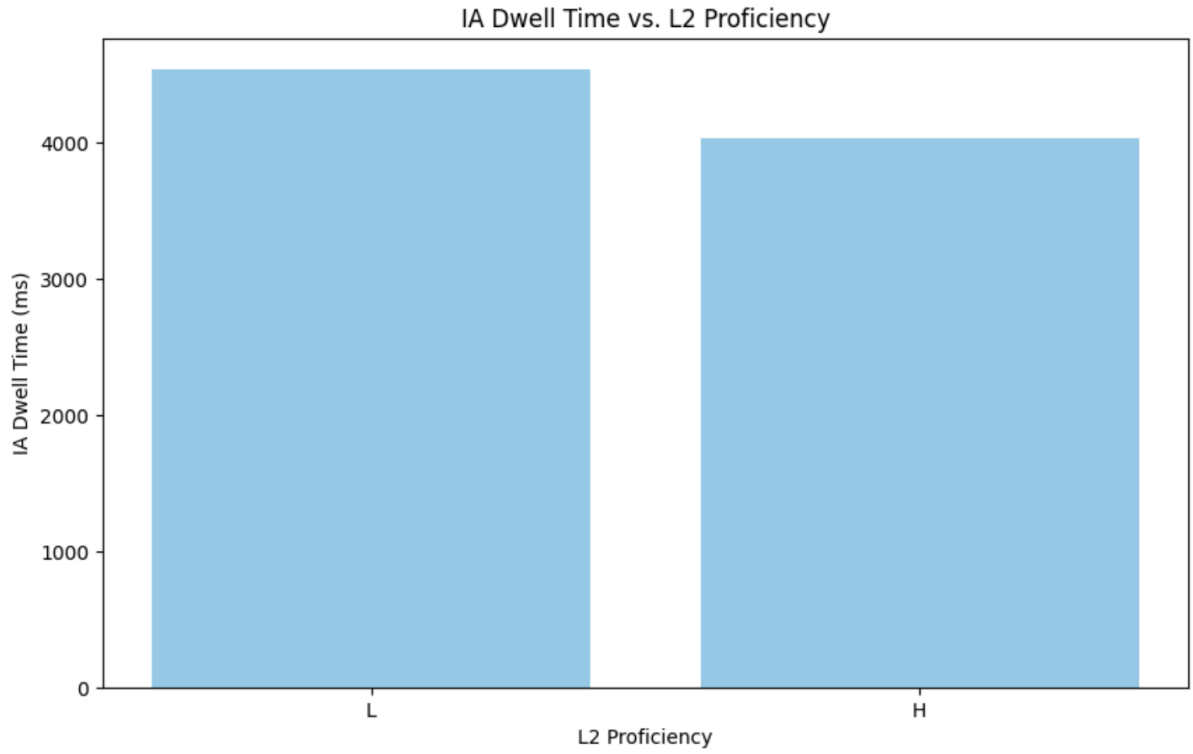
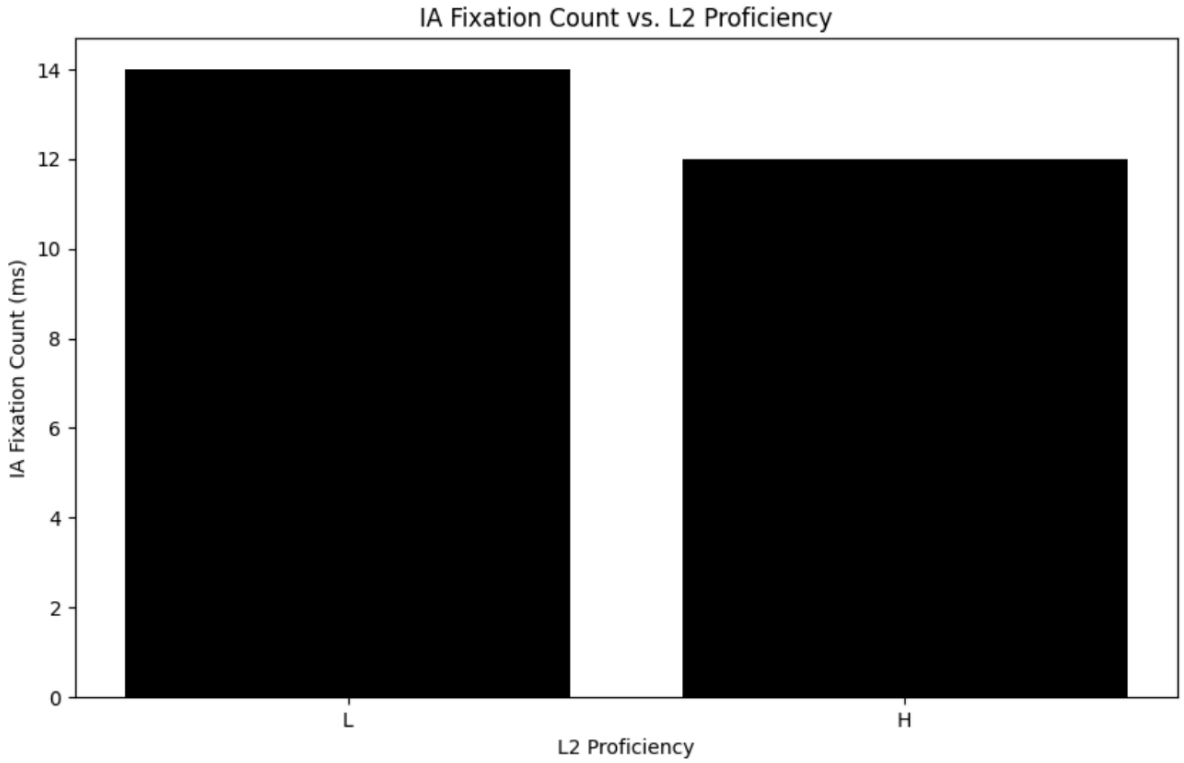
Column 1: Recording Session Label, Label of the data file of participants.
Column 2: L2 Proficiency, Participants' English proficiency (L represents Low, H represents High).
Column 3: Trial Index, Sequential order of the trial in the recording (from 1 to 160).
Column 4: Trial Label, Label of the trial, unique number for the sentences.
Column 5: Condition, Specific Conditions for each sentence (from Condition 1 to 8).
Column 6: IA_ID, Ordinal ID of the current interest area. ??
Column 7: IA_LABEL, Label for the visual form of each word for which the eye-movement measures are calculated.

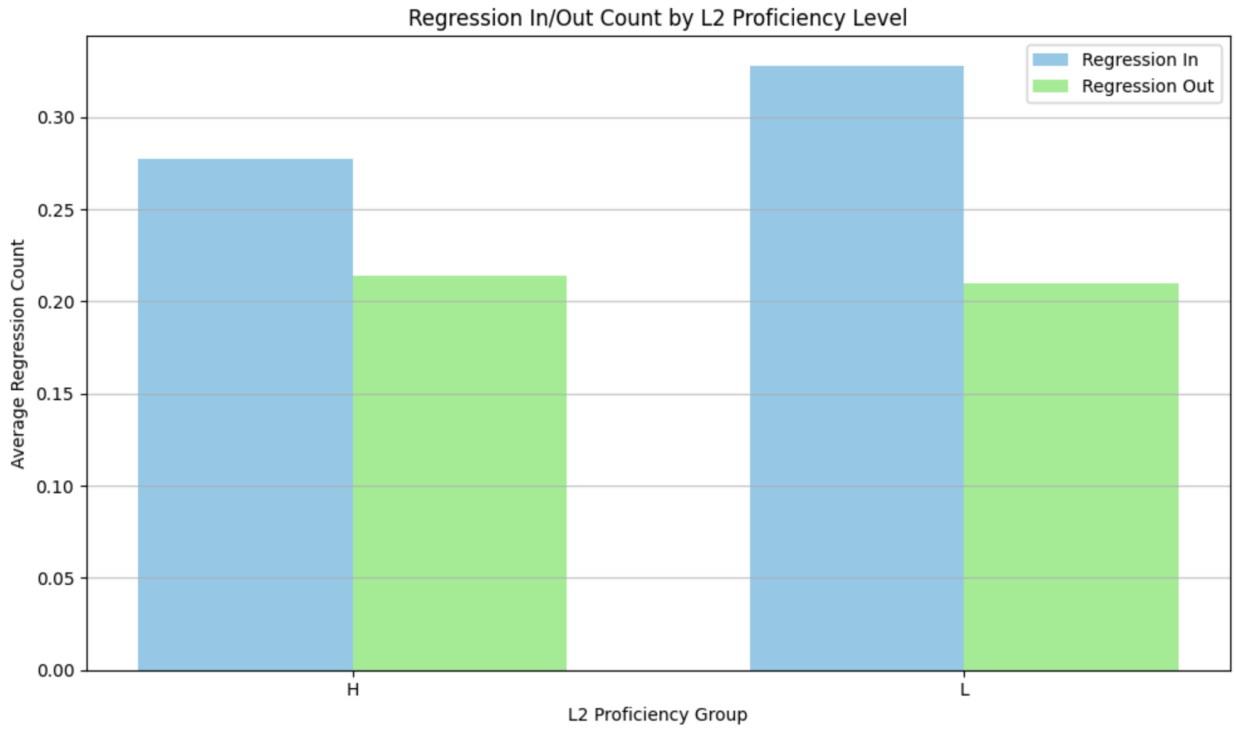
Features: (total 10, can be reduced for feature extraction)
Column 8, 9: IA First & Second Fixation Duration, provides the 1st & 2nd Fixation Duration (s) for each word.
Column 10: IA First Run Dwell Time*, provides the Gaze Duration for each word. (name)
Column 11: IA Regression Path Duration*, Records the total time spent on regression paths for each word. (**Selina)**
Column 12: IA Dwell Time, provides the total dwell time on each word <
Column 13: IA First Saccade Amplitude provides the angular distance the eyes travel during the initial saccadic movement when moving from one point of fixation to another for each word.
Column 14: IA Fixation Count*, Counts how long a participant's gaze fixates on each word (name) <
Column 15: IA Skip*, Provides the PS1 (Whether a word was skipped during the first pass of reading) for each word (1 = skipped, 0 = not skipped), (name)
Column 16: IA First Run Fixation Count, provides the 1st Fixation Count for each word.
Column 17: IA Regression In Count*, Counts how many times a reader's gaze regresses (returns/goes back) into a word while reading (name)
Column 18: IA Regression Out Count*, Counts how many times a reader's gaze regresses (moves their eyes backward) out of a word (name)

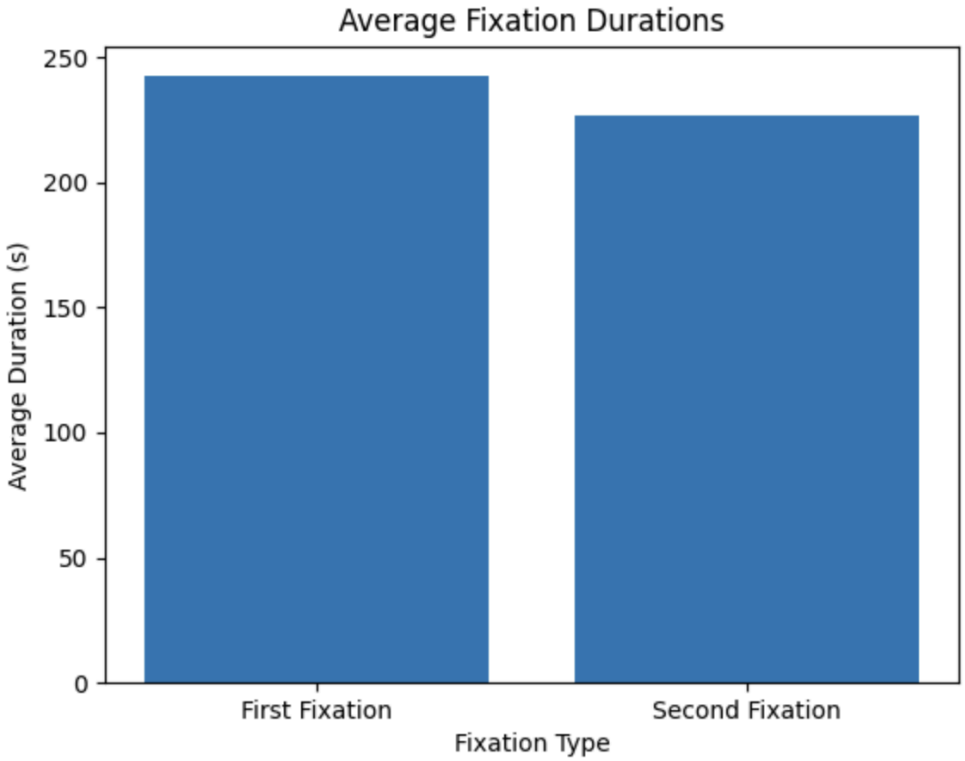
^^ lack of saccadic measurements, only have first (and second (removed due to missing data<)) saccade amplitude

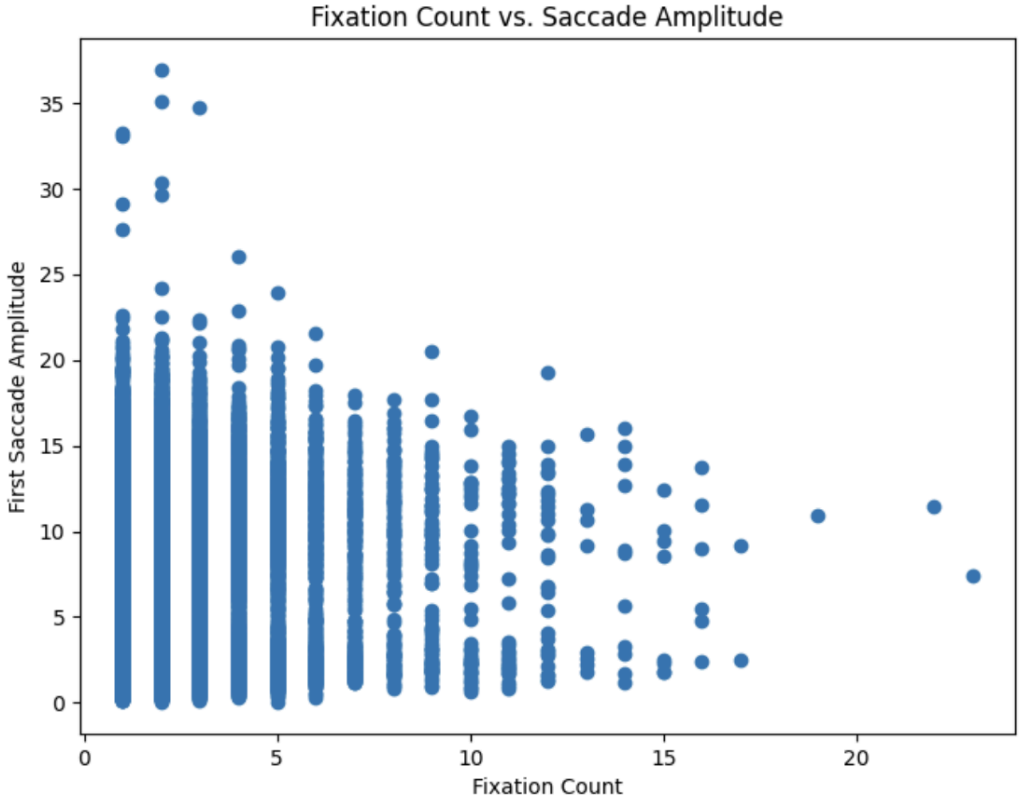
Insert Data Visualization graphs/ plots:






Regression In/Out Count by L2 Proficiency Level
Regression In
Regression Out
Average Regression Count
0.00
0.05
0.10
0.15
0.20
0.25
0.30
H
L
L2 Proficiency Group


Average Fixation Durations
Average Duration (s)
250
200
150
100
50
0
First Fixation
Second Fixation
Fixation Type

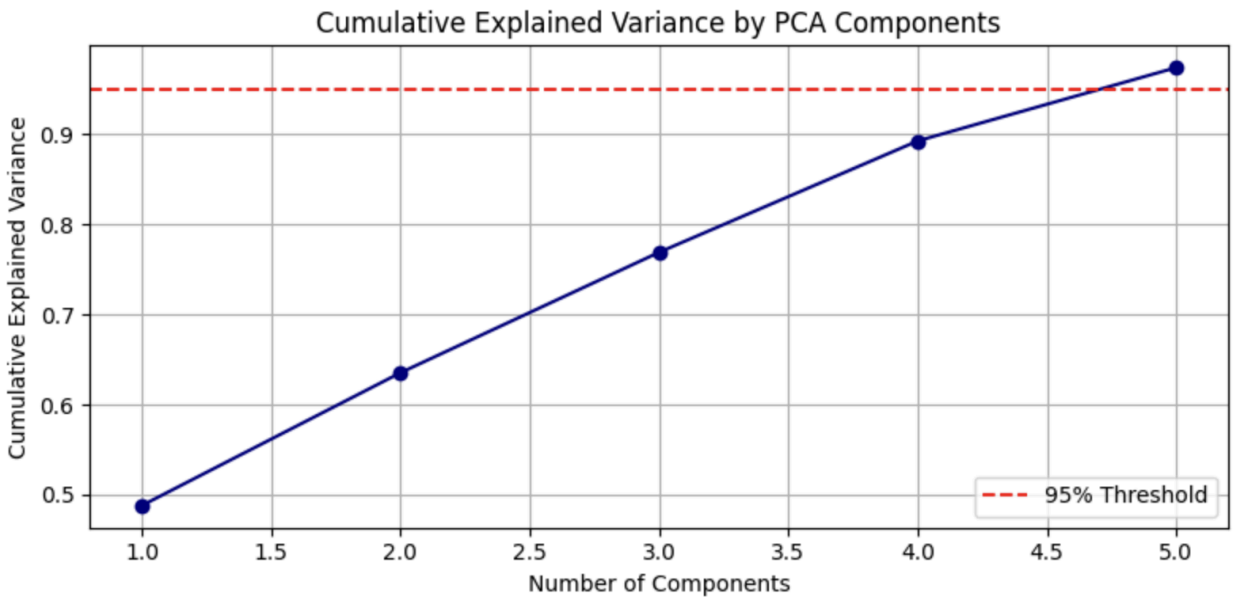

Fixation Count vs. Saccade Amplitude
First Saccade Amplitude
35
30
25
20
15
10
5
0
0
5
10
15
20
Fixation Count

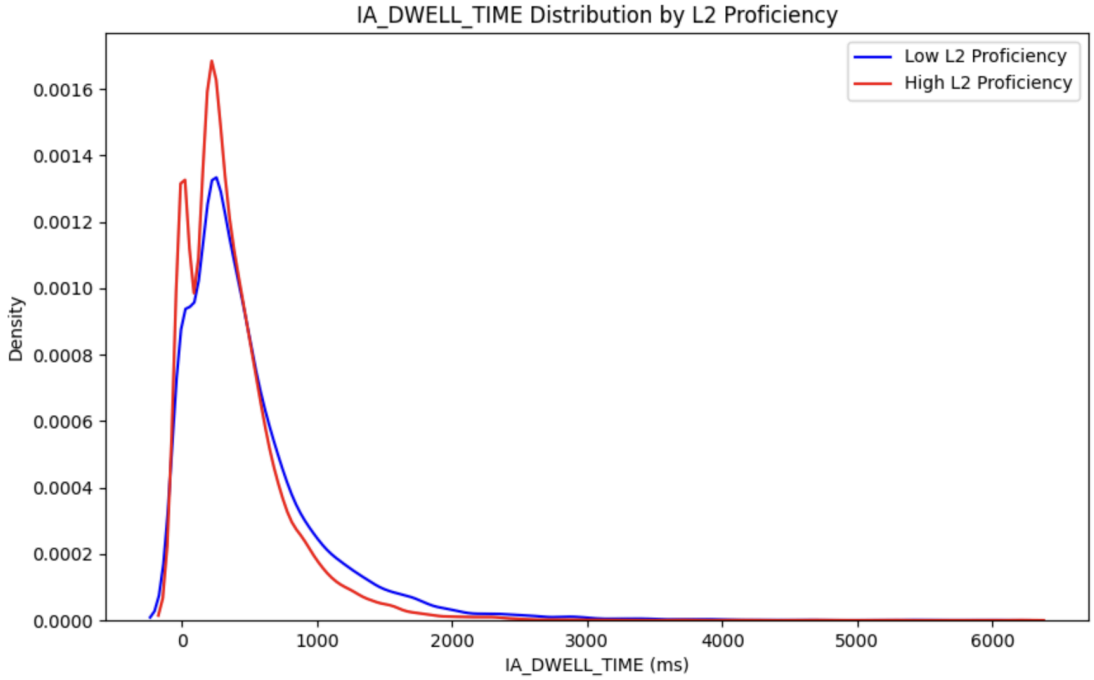
Top 100 Words by Mean Re-reading Effort

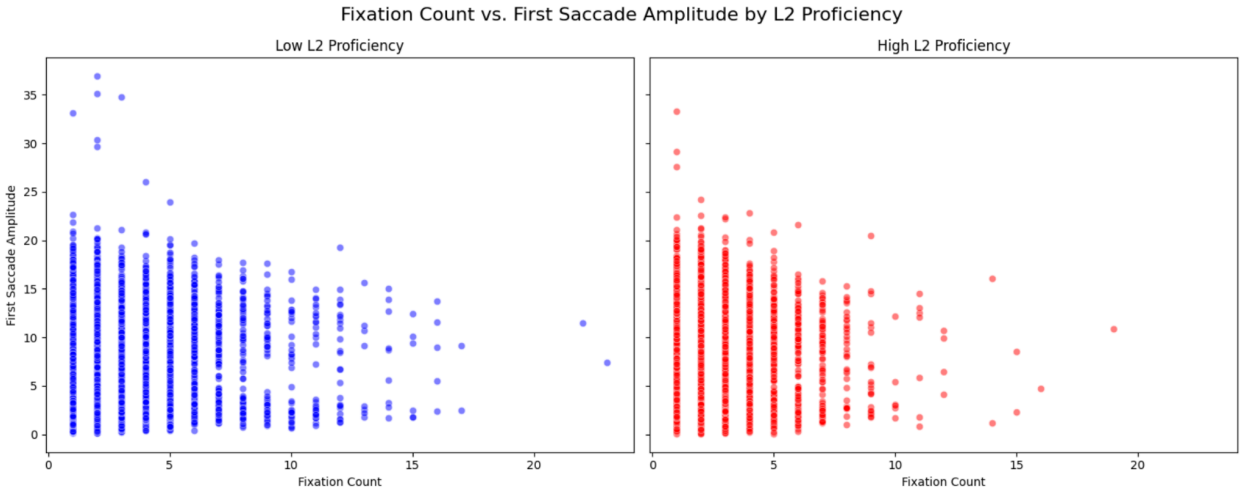
netizens	outstanding	insights	numerous	merchant	clown	careless	evidence	timid	shabby
diamond	captain	princess	kittens	cowards	bride	colleague	lovely	独到的	lawyer
sharpened	newborn	bachelors	president	负责的	soldiers	western	directors	landlord	building
necklace	planes	responsible	personal	version	generally	package	role	patient	academic
grassland	complex	owner	pirates	coaches	enemy	passenger	subway	wicked	second
破旧的	customers	phones	priest	passing	ancient	newspaper	judges	finance	brave
scholar	audience	naughty	ambitious	British	secretary	ambassador	proposal	famous	graduates
manager	guide	assistant	actions	generous	concrete	puppy	murderer	spies	scarves
fierce	scientist	thief	machines	journalists	photographer	designers	vast	excited	chickens
我们	大楼	picnic	cruel	waiters	citizens	greedy	buyers	defendant	players






Cumulative Explained Variance by PCA Components
Cumulative Explained Variance
0.5
0.6
0.7
0.8
0.9
1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5
5.0
Number of Components
95% Threshold


IA_DWELL_TIME Distribution by L2 Proficiency
Low L2 Proficiency
High L2 Proficiency
Density
0.0000
0.0002
0.0004
0.0006
0.0008
0.0010
0.0012
0.0014
0.0016
0
1000
2000
3000
4000
5000
6000
IA_DWELL_TIME (ms)


Fixation Count vs. First Saccade Amplitude by L2 Proficiency
Low L2 Proficiency
High L2 Proficiency
First Saccade Amplitude
Fixation Count
Fixation Count
0
5
10
15
20
25
30
35

Data Cleaning [preprocessing]

1. Missing Data

- Remove numeric features with missing data above 30%
- Eg. above 20% rn: SECOND_FIXATION_DURATION

2. **Drop rows missing critical columns**

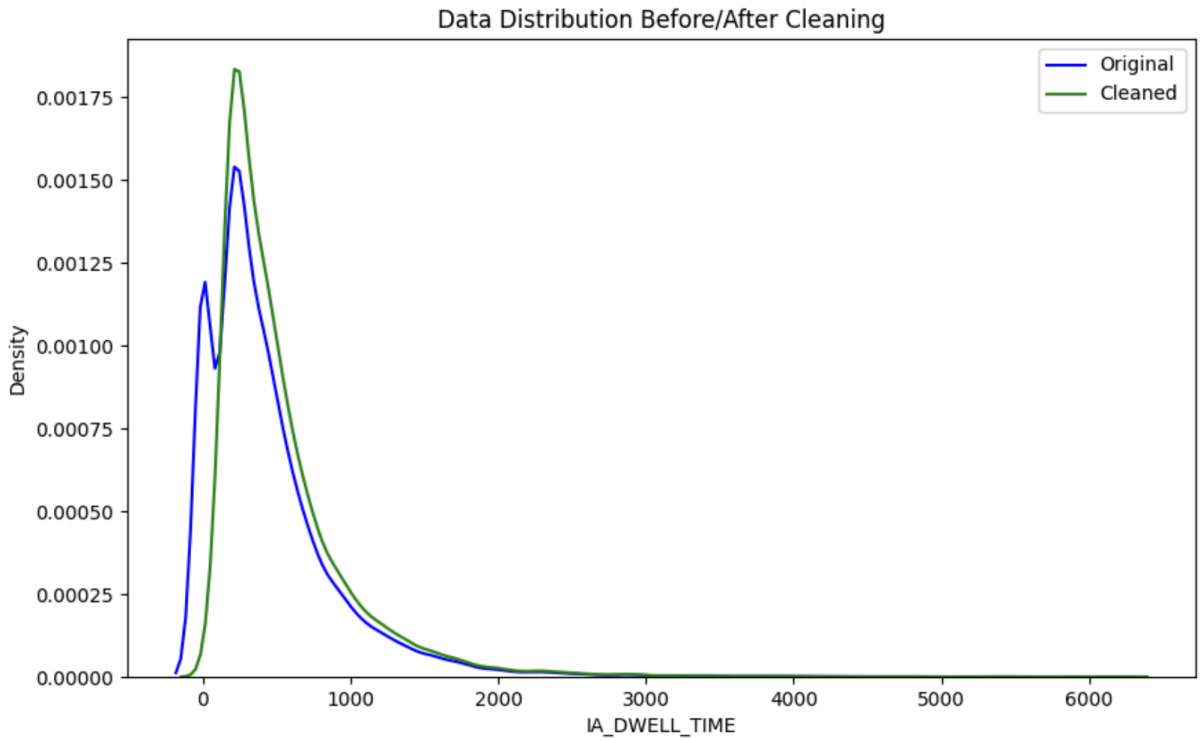
- Critical columns are those with a lot of N/A
- Critical: ['IA_REGRESSION_PATH_DURATION', 'IA_FIRST_FIXATION_DURATION', 'IA_FIRST_RUN_DWELL_TIME', 'IA_FIRST_SACCADE_AMPLITUDE']

3. **Log-Transforming Skewed Columns**

- Eye-tracking metrics (e.g., fixation durations, regressions) are often **log-normal**
- Applying np.log1p helps stabilize variance and improves model interpretability
- Auto-detects skew for skew > 1, does log-transformation
- **Skewed features**: ['IA_FIRST_FIXATION_DURATION', 'IA_FIRST_RUN_DWELL_TIME', 'IA_REGRESSION_PATH_DURATION', 'IA_DWELL_TIME', 'IA_FIXATION_COUNT', 'IA_SKIP', 'IA_FIRST_RUN_FIXATION_COUNT', 'IA_REGRESSION_IN_COUNT']

4. **Impute Non-Critical Numeric Columns with Median or Mean**

- If skewed → use **median**
- If not → use **mean**
- Replaces missing values with the mean or median of the column
-



5. **Min-Max Clipping of Extreme Outliers**

- Identifying Outliers: Z Score
    - Analyzes the Z-score of all data points and identifies outliers based on the parameter

    - Parameter: Z score > 3 standard deviations = outliers
    - E.g. found 443 outliers in regression path duration
- Cleaning Outliers: Clipping Numeric Feature Outliers
    - Clips very extreme outliers to the 1st and 99th percentiles to not losing rare valid data

6. **Drop Low-Variance Columns**

- If any numeric column has nearly no variance, it won't help your model.
- Drops low-variance columns

7. **Standardization**

- Method: Sklearn's StandardScaler
- Return standardized features for the model

1. Feature Engineering

1. Temporal Processing Features

Fixation Duration Ratios: Calculate the ratio between first and second fixation durations (Column 8/Column 9) to capture processing difficulty changes

Processing Time Differences: Create features that measure the difference between total dwell time (Column 12) and first run dwell time (Column 10) to quantify re-reading behavior

Normalized Duration Metrics: Scale fixation durations by word length to account for word-length effects

Fixation Density Differences: Measures of how fixation patterns differ between proficiency levels

2. Regression-Based Features

Regression Complexity Index: Combine regression in/out counts (Columns 17-18) with regression path duration (Column 11) to create a composite measure of processing difficulty

Regression Proportion: Calculate the proportion of total reading time spent in regressions (Column 11/Column 12)

Regression Density: Create a feature representing regressions per fixation (Column 17/Column 14)

3. Saccade-Related Features

Saccade Efficiency Ratio: Calculate the ratio between saccade amplitude (Column 13) and subsequent fixation duration to capture processing efficiency

Saccade Amplitude Variability: Compute the moving standard deviation of saccade amplitudes across sequential words

Forward/Backward Movement Ratio: Create a feature representing the ratio of progressive to regressive eye movements

4. Sequence-Based Features
Fixation Sequence Patterns: Create n-gram features of fixation sequences (e.g., skip followed by regression) ???

Reading Rhythm Disruptions: Identify points where the typical reading rhythm changes (sudden increases in fixation duration or regression)

5. Word-Skipping Features
Skip Patterns: Create features representing patterns of skipped and non-skipped words (Column 15)

Skip-Regression Relationship: Create interaction features between skipping behavior and subsequent regressions

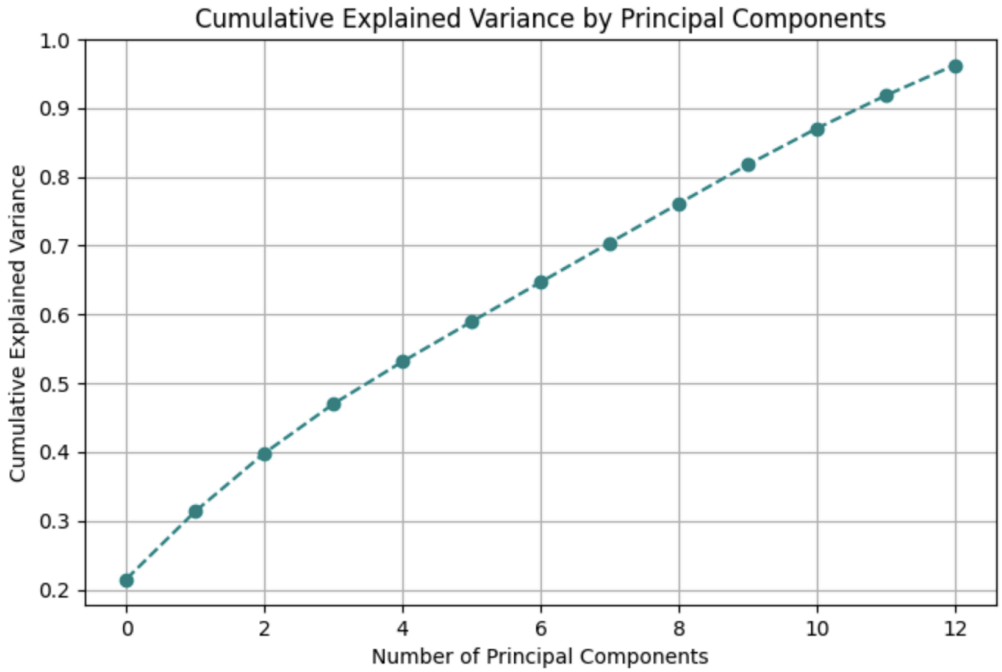
Skip Recovery Cost: Measure fixation durations following skipped words

6. Contextual Window Features
Moving Averages: Calculate moving averages of key metrics (fixation durations, regression counts) using different window sizes

Local Maxima Detection: Identify peaks in fixation durations or regression counts that might indicate processing difficulty

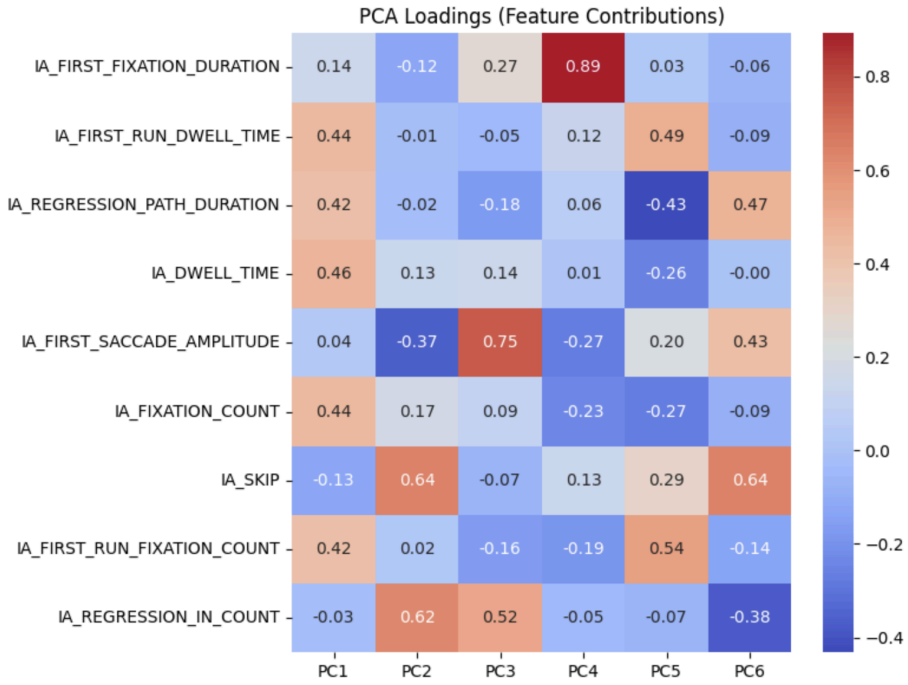
Pre-Switch Reading Patterns: Create features specifically focused on reading patterns immediately before potential code-switch points

Principal Component Analysis:

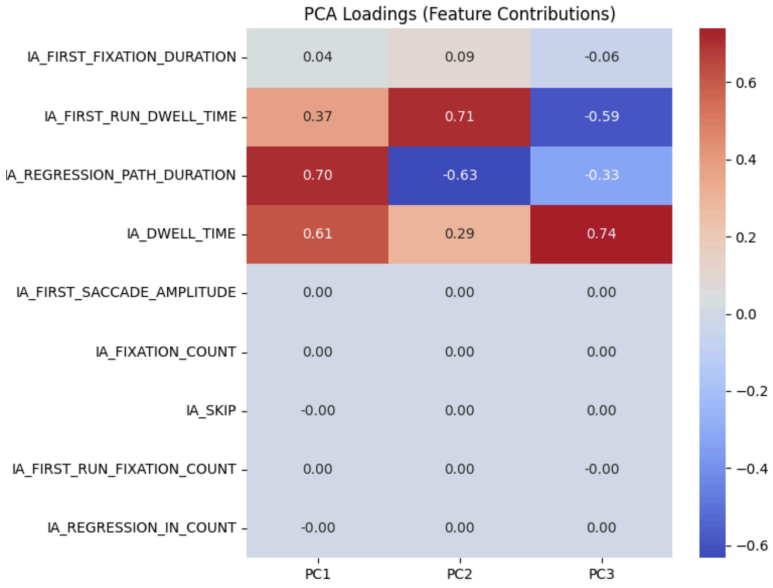


Loadings Heat Map

- PC5 – strong correlation with First Fixation Duration → 0.89
- PC4 – strong correlation with First Saccade Amplitude → 0.75
- PC3 – correlation with Regression In Count & IA Skip → 0.60, 0.62
- PC1 – correlation with Dwell Time, Fixation Count, First Run Dwell Time, Regression Path Duration, First Run Fixation Count (0.46, 0.44, 0.43, 0.42, 0.41)
- PC2 – correlated with Language Proficiency (+ high, - low)
- PC6 – correlated with First Run Fixation Count + First Run Dwell Time → 0.54, 0.49
- PC7 – correlated with IA Skip → 0.64



Updated PCA below: (means we only need to use 3 features to predict code-switching)



Correlation Analysis

2. Feature Selection

a) Correlation Analysis

- Paper uses Spearman's rank correlation between features to identify which ones provided similar information.

- After identifying highly correlated features, you can select one representative from each correlated group for your model, which helps reduce dimensionality while preserving most of the information.

**remove Filler ones

a. further normalize the data, try to print out the data after the normalization part to visually inspect those for accuracy (prior to feeding them to SVM), check for N/A, repeated values and 0s
b. shrink the dataset, rather use 20% of the whole dataset but with very clean data entries than using the entire dataset, focus on some features only <<
c. generate descriptive statistics for variables (mean, standard deviation, t-test to find the significant difference…)

Either remove

- Imputation method only mean
- Remove skew log transform
-

Models Tried: SVM, Random Forests, Gradient Boost Estimator, Logistic Regression
Code-Switching Prediction: Labels on each word's eye-tracking data

- 0 = no code switching
- 1 = pre-code switching (eg, original language)
- 2 = after code switching (into another/second language)

Model: Gradient Boost Estimator

Problems:

- Class imbalance in the test set (e.g., 2300 vs 1800): not terrible, but the model might be over-prioritizing class 1.
- Feature overlap or insufficient discriminative signal for class 0.
- Too much noise in class 0 samples even after cleaning.

Class Imbalance Strategies: SMOTE

Model: SVM
Selina's initial results:
SVM Accuracy: 0.4945
SVM F1 Score: 0.4290

SVM Classification Report:

	precision	recall	f1-score	support
0	0.33	0.02	0.03	1767
1	0.49	0.75	0.59	2818
2	0.51	0.54	0.53	2565
accuracy			0.49	7150
macro avg	0.44	0.44	0.38	7150
weighted avg	0.46	0.49	0.43	7150

Hyperparameters: kernel='rbf', C=10, gamma='scale', probability=True, random_state=42

Model #2: Random Forests

Selina's initial results:
Random Forest Accuracy: 0.4750
Random Forest F1 Score: 0.4321

Random Forest Classification Report:

	precision	recall	f1-score	support
0	0.26	0.07	0.11	1767

1	0.49	0.72	0.58	2818
2	0.49	0.49	0.49	2565
accuracy			0.47	7150
macro avg	0.42	0.43	0.39	7150
weighted avg	0.43	0.47	0.43	7150

Hyperparameters: n_estimators=500, random_state=42, max_depth=70, min_samples_split=10, min_samples_leaf=4

Useful sources:
https://medium.com/@ap.nattapoj_st/basic-comparison-between-randomforest-svm-and-xgboost-0e5862871175
https://bmcbioinformatics.biomedcentral.com/articles/10.1186/1471-2105-9-319
https://towardsdatascience.com/hyperparameter-tuning-the-random-forest-in-python-using-scikit-learn-28d2aa77dd74/
https://pmc.ncbi.nlm.nih.gov/articles/PMC9299283/
https://f0nzie.github.io/machine_learning_compilation/tuning-of-support-vector-machine-prediction.html
https://math.ucsd.edu/sites/math.ucsd.edu/files/undergrad/honors-program/honors-theses/2020-2021/Huanxi_Liu_Honors_Thesis.pdf